

Financial Fraud Detection System

Project Report — Executive Summary & Results

Project Name: Financial Fraud Detection Model & Streamlit Dashboard

Technologies: Python, Streamlit, Scikit-Learn, XGBoost, Pandas, Plotly, NumPy

Status: Completed & Deployed (local dashboard at <http://localhost:8501>)

1. Executive Summary

Financial fraud presents a multi-billion-dollar risk for digital banking ecosystems, spanning credit cards, debit cards, UPI, NetBanking, and PayPal channels. This project delivers an end-to-end machine learning and analytics system that identifies fraudulent transactions in real time, computes risk-probability scores, and equips fraud analysts with actionable, explainable insights through an interactive dashboard.

Three candidate models were benchmarked, and Logistic Regression was selected for deployment on the strength of its 100% recall on the test set — meaning zero fraudulent transactions were missed.

2. Business Problem & Objective

Fraudulent transactions represent a small but highly costly fraction of overall volume (approximately 9.6% of the training data). Because a missed fraud case (false negative) carries far greater financial and reputational cost than a false alarm (false positive), the objective of this project was to build a system optimized to catch as much fraud as possible, while remaining operationally practical for analysts to act on.

The system was built to detect five recognized fraud patterns:

- Card-Not-Present (CNP) fraud in online transactions
- Behavioral anomaly fraud — spending that sharply diverges from a customer's baseline
- Cross-border / international fraud from high-risk, unnotified locations
- Circadian (night-time) exploitation, e.g. transactions between 11 PM and 5 AM
- Suspicious-keyword fraud flagged via transaction remarks

3. Methodology at a Glance

Raw transaction data was cleaned (missing-value imputation, outlier detection via IQR and Z-score methods, and feature scaling) and enriched with four engineered risk indicators, including a spending-surge ratio, absolute spend variance, a night-transaction flag, and a suspicious-keyword indicator. Three classification algorithms — Logistic Regression, Random Forest, and XGBoost — were trained and benchmarked on this enriched feature set.

4. Model Performance Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
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Logistic Regression (Selected)	75.40%	28.07%	100.00%	0.4384	0.8933
Random Forest Classifier	77.20%	26.26%	76.04%	0.3904	0.8580
XGBoost Classifier	83.30%	28.22%	47.92%	0.3552	0.8665

Key Finding: Logistic Regression was selected as the production model because it achieved 100% recall (zero missed fraud cases) and the highest ROC-AUC score of 0.8933 among the three models tested. While XGBoost scored higher on raw accuracy, its lower recall (47.92%) would let over half of all fraud cases through undetected — an unacceptable trade-off given the asymmetric cost of missed fraud.

5. Dashboard Overview

The system is delivered as a five-module interactive Streamlit web application:

- Executive Dashboard — real-time KPI cards (total volume, fraud rate, flagged loss) and visual breakdowns by payment method, merchant category, location, and hour of day
- Real-Time Fraud Predictor — manual transaction entry with an instant risk score and a plain-language risk badge (High / Moderate / Safe)
- Analytics & Insights — behavioral anomaly scatter plots, device-type risk rates, and suspicious-keyword correlation views
- Model Performance — side-by-side benchmark comparison and top-10 influential feature ranking
- Batch CSV Scanner — drag-and-drop bulk scanning with automatic column mapping and a downloadable, labeled results file

6. Business Impact & Recommendations

- Deploy the Logistic Regression model in production for its zero-miss recall profile on validated data
- Route all high-risk flags to manual analyst review rather than automatic decline, given the current false-positive rate
- Periodically retrain on fresh transaction data to maintain detection accuracy as fraud patterns evolve
- Extend the feature set with device-fingerprinting and velocity-based signals in a future iteration to improve precision

7. Conclusion

This project demonstrates a complete, production-ready pipeline — from raw transaction data through feature engineering, model benchmarking, and an operational analyst dashboard. The selected Logistic Regression model provides the strongest safety net against missed fraud, and the accompanying dashboard gives fraud teams the visibility and tools needed to act on its predictions quickly.

Appendix A — Quick Start Commands

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# Launch the interactive dashboard
streamlit run app.py

# Re-train the ML model
python train_model.py
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